Stat 159/259: Reproducible & Collaborative Statistical Data Science

Lab 01: Introductions, DataHub & the Shell

Today’s plan

01	**Introductions**	Meet your GSI and your neighbors
02	**The course’s Hubs: DataHub, GitHub**	Get your environment up and running in the browser
03	**Basic Shell Commands**	pwd, ls, cd, who, date, cal, man
04	**Navigating Files & Directories**	mkdir, rmdir, touch, cp, rm, mv, wc
05	**Hands-On Practice**	Work through the notebook

Introductions

About me…

My name is Larissa (she/her)

- Statistics Master’s program (3rd semester)
- I’m from Mexico.
- I don’t like matcha and I love my dogs back home
- Office Hours:
  - Mondays and Thursdays - 12 pm - 2 pm
- e-mail: larissa_arreola@berkeley.edu

Introductions

About me…

- ➢ Sequoia Andrade (she/her) (srandrade@berkeley.edu)
- ➢ 3rd year PhD Candidate in Statistics
  - ○ Computational statistics for applications in carbon monitoring and social justice
- ➢ Originally from the Santa Cruz mountains (CA)
- ➢ Spend most of my free time with my cats, swimming, lifting, playing softball, baking, and caring for my plants

Meet your neighbors

Introduce yourselves:

- ➢ Name, program, one thing you don’t like
- ➢ In teams, you’ll have 45 seconds to guess each image for the trivia.

Lab logistics

1. This is a hands-on class so lab attendance is mandatory!
2. We start on Berkeley time
3. 50 minutes - break - 50 minutes roughly
4. There will be a reading quiz towards the end of discussion. Super short and easy (if you read the material)
5. Quizzes will not be graded for this lab and next (beginning Week 3)
6. Mostly working in pairs or groups on coding/software exercises, class discussion, working on HW assignments
7. Jupyter Notebooks/rmd notebooks with lab material will always be posted on course website under the lab tabs

Our work environment: **DataHub**

We all have the exact same setup and we don't have to install anything locally.

The link: stat159.datahub.berkeley.edu

(Make sure to keep it somewhere handy because we will be using it all semester long)

GitHub

We will handle your course submissions through GitHub, so it's very important that you have an account!

Do you have an account?

Yes

And I have shared my username in the intro survey → Perfect, enjoy a 5 min break!

But I didn't share my username in the intro survey → Please, come with me to write it down

No :(

Go to github.com and create one right now! Then come with me and share your username

The Terminal (or Shell)

There are usually 2 ways to interact with the computer through the GUI or through the CLI

- ➔ **GUI (Graphical User Interface):** you interact with a computer by pointing, clicking, and dragging (windows, icons, buttons, menus).

- ➔ **CLI (Command Line Interface):** you interact with a computer by typing text commands into a terminal, and the computer talks back in text. No mouse required. (Like hackers in the movies)

(Why) the Terminal

We are aiming towards generalizable work products. One of the first questions we can ask is **how can we generalize/repeat/reproduce our communication across different machines?**

> by navigating your computer through the terminal (*we are focusing in the UNIX terminal)

01
Reproducibility

Typed commands are explicit, precise, and re-runnable.

02
Automation

We can chain commands into scripts that run the same way every time.

03
Foundation for other tools

Version control, servers, and data pipelines all speak shell.

Basic Commands at a glance

pwd — Print Working Directory — where am I?

`$ pwd`

ls — List the contents of a directory

`$ ls -la`

cd — Change Directory — move around

`$ cd notebooks/`

who — Show who is logged in to the system

`$ who`

date — Show the current date and time

`$ date`

cal — Display a calendar in the terminal

`$ cal`

man — Manual — read the built-in help pages for any command

`$ man ls` → `press q to quit`

Image source: Claude

Navigating Files & Directories

mkdir — Make a new directory (folder)

`$ mkdir data`

rmdir — Remove an empty directory

`$ rmdir old_data`

touch — Create a new, empty file

`$ touch notes.txt`

cp — Copy a file or directory

`$ cp a.txt b.txt`

rm — Remove (delete) a file — careful, no undo!

`$ rm draft.txt`

mv — Move or rename a file/directory

`$ mv old.txt new.txt`

wc — Word Count — counts lines, words, and characters in a file

`$ wc -l data.csv`

Image source: Claude

Let’s try them together

1. **Open DataHub -> then the terminal**
2. Find out where you are standing (what is the command for it?)
3. Let’s create a stat159 folder (with mkdir)
4. Go into the stat159 folder (with cd)
5. Create three new folders: *labs*, *demo* & *homeworks* (you can create more than one folder at a time with mkdir)
6. Now create an empty txt file (with touch) - call it *notes.txt*

Let’s try them together

7. List what you have inside the stat159 folder (with ls)
8. Copy notes.txt into the *homeworks* folder (remember where you are), renaming it *notes_copy.txt* (with cp)
9. Move the original *notes.txt* file into the labs folder (with mv)
10. List what you have again
11. Go into both *labs* & *homework* to see if the empty files are there
12. Remove the empty files from the folders (with rm)
13. Finally remove the *demo* folder.

Basic Commands + Flags

A **flag** (or "option" or "switch") modifies how a command behaves. Without flags, most commands just do their simplest possible thing.

General syntax:

command [flags] [arguments]

Short vs. long flags

- **Short flags** start with a single dash and are usually one letter: -l, -a, -r
- **Long flags** start with two dashes and are more descriptive: --all, --recursive, --verbose

Combining short flags

You can usually stack short flags together behind a single dash:

```
ls -l -a          # two separate flags
ls -la            # same thing, combined
ls -al            # order usually doesn't matter
```

*check the **man** page if you need help

Your time to practice with the notebooks

➔ Download the notebook file (lab01_shell_practice.ipynb) from bCourses*, follow the instructions. Let me know if you have any questions.

*For those of you who don't have access to bCourses, let me know now so I can add you right now!

Questions?

See you next week :)